

Day 21 (Part 4): BPDU Guard & BPDU Filter (STP Toolkit)

PortFast & BPDUs

```
SW1(config)# interface g0/1
SW1(config-if)# spanning-tree portfast
SW1(config-if)# do show spanning-tree interface g0/1 detail
!output omitted
The port is in the portfast edge mode
Link type is point-to-point by default
BPDU: sent 77, received 0
```

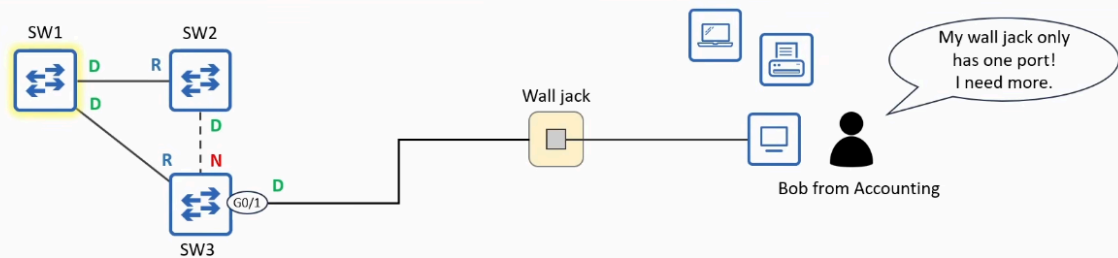
- **PortFast** makes a port start in the Forwarding state when it is connected, but it doesn't disable STP on the port.
 - The port will continue to send BPDUs every 2 seconds.
- Because end hosts don't run STP and send BPDUs, a PortFast-enabled port shouldn't receive BPDUs.
 - But what if it does?

PortFast & BPDUs

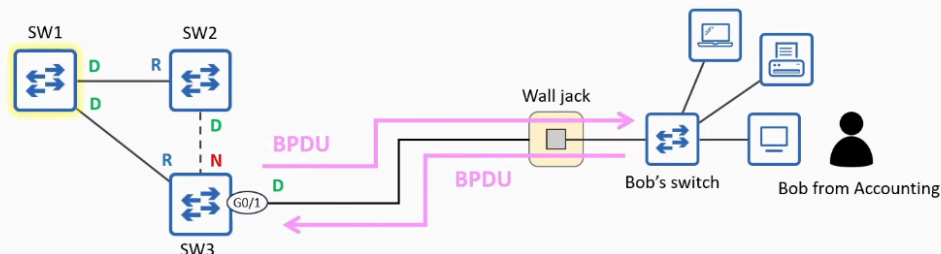
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 - The port will continue to send BPDUs every 2 seconds.
- Because end hosts don't run STP and send BPDUs, a PortFast-enabled port shouldn't receive BPDUs.
 - But what if it does?
- If a PortFast-enabled port receives an STP BPDU, it will revert to acting like a regular STP port (without PortFast).

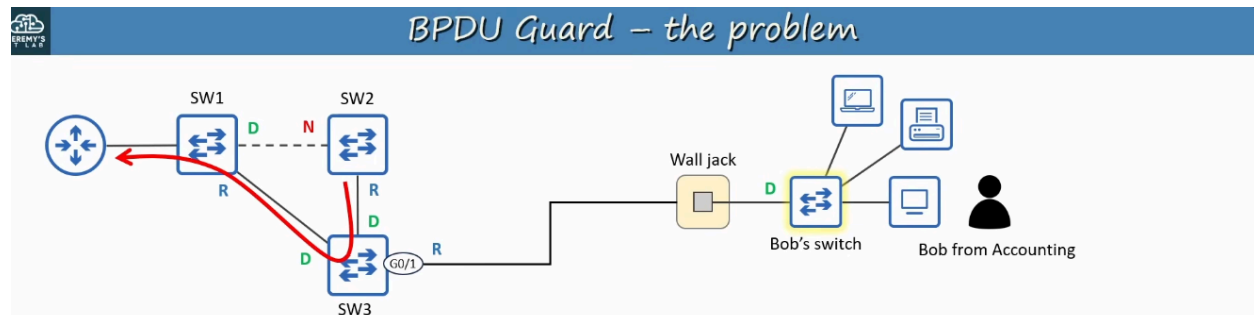
BPDUs Guard – the problem



BPDUs Guard – the problem



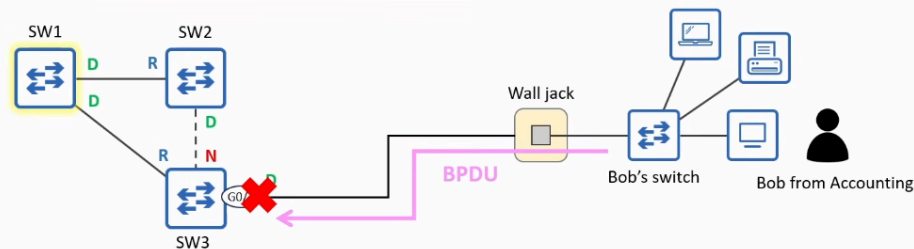
If Bob's switch becomes Root Bridge



- PortFast should only be enabled on ports connected to non-switch devices (end hosts, routers).
 - These devices don't send BPDUs.
- A PortFast-enabled port still sends BPDUs and will operate like a regular STP port if it receives BPDUs from a neighbor.
- If an end user carelessly connects a switch to a port meant for end hosts, it could affect the STP topology.
 - **BPDUs Guard** acts as a safeguard against this.



BPDU Guard – the solution



- **BPDU Guard** protects the network from unauthorized switches being connected to ports intended for end hosts.
- It can be configured separately from **PortFast**, but both features are usually used together.
 - They both enhance STP's functionality on ports intended for end hosts.
- A BPDU Guard-enabled port continues to send BPDUs, but if it receives a BPDU it enters the **error-disabled** state.
 - In effect, this disables the port.



BPDU Guard configuration

- Like **PortFast**, **BPDU Guard** can be configured in two ways:

- Per-port:

```
SW3(config)# interface g0/1
SW3(config-if)# spanning-tree bpduguard enable
SW3(config-if)# do show spanning-tree interface g0/1 detail
Port 2 (GigabitEthernet0/1) of VLAN0001 is designated forwarding
  Port path cost 4, Port priority 128, Port Identifier 128.2.
  Designated root has priority 24577, address 5254.001a.9d29
  Designated bridge has priority 32769, address 5254.0006.448f
  Designated port id is 128.2, designated path cost 4
  Timers: message age 0, forward delay 0, hold 0
  Number of transitions to forwarding state: 1
  The port is in the portfast edge mode
  Link type is point-to-point by default
  Bpdu guard is enabled
  BPDU: sent 93, received 0
```

- Default:

```
SW3(config)# spanning-tree portfast bpduguard default
SW3(config)# do show spanning-tree interface g0/1 detail
Port 2 (GigabitEthernet0/1) of VLAN0001 is designated forwarding
  Port path cost 4, Port priority 128, Port Identifier 128.2.
  Designated root has priority 24577, address 5254.001a.9d29
  Designated bridge has priority 32769, address 5254.0006.448f
  Designated port id is 128.2, designated path cost 4
  Timers: message age 0, forward delay 0, hold 0
  Number of transitions to forwarding state: 1
  The port is in the portfast edge mode
  Link type is point-to-point by default
  Bpdu guard is enabled by default
  BPDU: sent 165, received 0
```

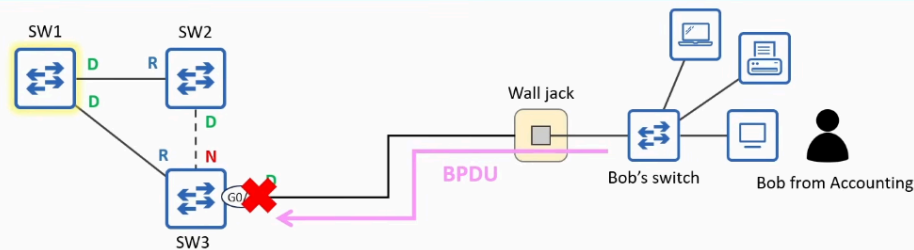
spanning-tree portfast [edge] bpduguard default

- When enabled by default, **BPDU Guard** is activated on all Portfast-enabled ports.
- Use **spanning-tree bpduguard disable** in interface config mode to disable it on specific ports.





ErrDisable

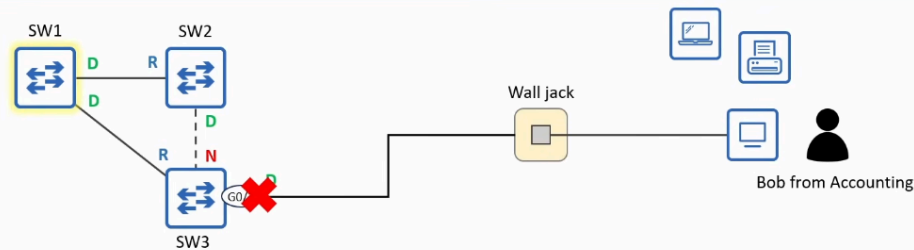


```
*Sep 3 05:08:11.977: %SPANTREE-2-BLOCK_BPDUGUARD: Received BPDU on port GigabitEthernet0/1 with BPDU Guard enabled. Disabling port.
*Sep 3 05:08:11.977: %PM-4-ERR_DISABLE: bpduguard error detected on Gi0/1, putting Gi0/1 in err-disable state
*Sep 3 05:08:12.978: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
*Sep 3 05:08:13.978: %LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to down
SW3# show interfaces g0/1
GigabitEthernet0/1 is down, line protocol is down (err-disabled)
!output omitted
```

- **ErrDisable** is a Cisco switch feature that disables a port under certain conditions, such as a BPDU Guard violation.
 - You will learn a few others for the CCNA exam, such as:
 - Power Policing violations
 - Port Security violations
 - DAI (Dynamic ARP Inspection) violations



ErrDisable



- To re-enable an err-disabled port, first solve the underlying issue.
 - If you re-enable the port without fixing the issue, it will just be err-disabled again.
- You can re-enable an err-disabled port in two ways:
 1. Manual: use **shutdown** and **no shutdown** to reset the disabled port.
 2. Automatic: **ErrDisable Recovery**



ErrDisable Recovery

- **ErrDisable Recovery** is a feature that automatically re-enables err-disabled ports after a certain period of time.

```
SW3# show errdisable recovery
```

```
ErrDisable Reason      Timer Status
-----
arp-inspection          Disabled
bpduguard               Disabled
channel-misconfig (STP) Disabled
!output omitted
```

```
Timer interval: 300 seconds
```

```
Interfaces that will be enabled at the next timeout:
```

ErrDisable Recovery is disabled by default.

- The default recovery timer is 300 seconds (5 minutes).
- Err-disabled interfaces will be automatically re-enabled after 5 minutes.
- Use `SW1(config)# errdisable recovery interval seconds` to modify the interval.

- Use `errdisable recovery cause cause` to enable ErrDisable Recovery for ports disabled by a particular cause.

```
SW3(config)# errdisable recovery cause bpduguard
```

```
SW3(config)# do show errdisable recovery
```

```
ErrDisable Reason      Timer Status
-----
arp-inspection          Disabled
bpduguard               Enabled
channel-misconfig (STP) Disabled
```

```
Timer interval: 300 seconds
```

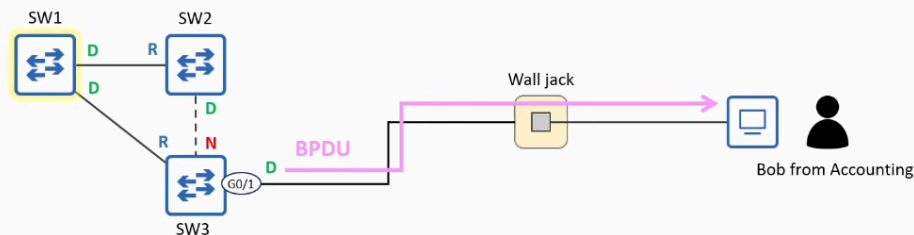
```
Interfaces that will be enabled at the next timeout:
```

```
Interface      Errdisable reason  Time left(sec)
-----
Gi0/1          bpduguard          296
```

Gi0/1 will be automatically re-enabled after 296 seconds.



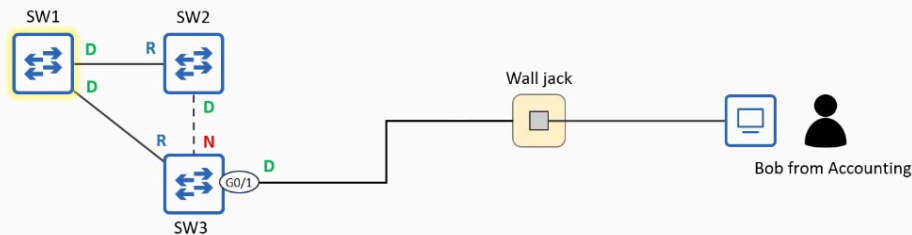
BPDU Filter – the problem



- A switch port connected to an end host continues sending BPDUs every 2 seconds.
 - regardless of whether PortFast and/or BPDU Guard are enabled
- If the port doesn't connect to a switch, sending BPDUs is unnecessary and undesirable for a couple of reasons:
 1. Sending BPDUs uses some bandwidth and processing power on the switch (although it's minimal).
 2. BPDUs contain information about the LAN's STP topology.
 - If maximum security is a concern, you should avoid sending this info to user devices.
- **BPDU Filter** solves this by preventing a port from sending BPDUs.



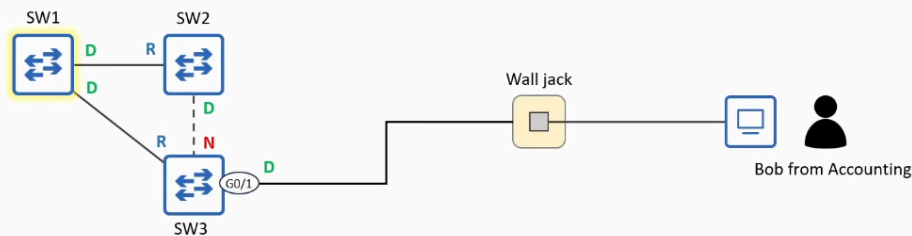
BPDUs Filter – the solution



- **BPDUs Filter** stops a port from sending BPDUs.
 - Unlike BPDUs Guard, it does not disable the port if it receives a BPDU.
- BPDUs Filter can be enabled in two ways:
 - Per-port: SW3(config-if)# **spanning-tree bpdudfilter enable**
 - The port will not send BPDUs.
 - The port will ignore any BPDUs it receives.
 - In effect, this disables STP on the port. **Use with caution!**
 - Default: SW3(config)# **spanning-tree portfast [edge] bpdudfilter default**
 - BPDUs Filter will be activated on all PortFast-enabled ports.
 - You can use **spanning-tree bpdudfilter disable** to disable it on specific ports.
 - The port will not send BPDUs.
 - If the port receives a BPDU, PortFast and BPDUs Filter are disabled, and it operates as a normal STP port.



BPDUs Filter – the solution



- My recommendation:
 - Enable PortFast and BPDUs Guard however you prefer (per-port or by default).
 - Only enable BPDUs Filter by default (global config mode).
 - unless you have a very good reason to enable it per-port

BPDUs Guard and BPDUs Filter can be enabled on the same port at the same time:

- If BPDUs Filter is enabled in global config mode and the port receives a BPDU:
 1. BPDUs Filter will be disabled.
 2. BPDUs Guard will be triggered (and err-disable the interface).
- If BPDUs Filter is enabled in interface config mode and the port receives a BPDU:
 - The BPDU will be ignored.
 - BPDUs Guard will **not** be triggered.





Summary

- **PortFast** should only be enabled on ports connected to non-switch devices (end hosts, routers) that don't send BPDUs.
 - A PortFast-enabled port still sends BPDUs and will operate like a regular STP port if it receives BPDUs from a neighbor.
 - If an end user carelessly connects a switch to a port meant for end hosts, it could affect the STP topology
- **BPDU Guard** protects the network from unauthorized switches being connected to ports intended for end hosts.
 - If the port receives a BPDU, it enters the **error-disabled** (err-disabled) state, effectively disabling the port.
 - Per-port: SW1(config-if)# **spanning-tree bpduguard enable**
 - Default: SW1(config)# **spanning-tree portfast [edge] bpduguard default**
 - Enables BPDU Guard on all PortFast-enabled ports.
 - Use **spanning-tree bpduguard disable** to disable it on specific ports.
- An err-disabled port can be re-enabled in two ways:
 1. Manual: **shutdown** and **no shutdown**
 2. Automatic: **ErrDisable Recovery**
 - SW1(config)# **errdisable recovery cause bpduguard**
 - In either case, make sure you fix the underlying problem that caused the port to be err-disabled.
- **BPDU Filter** prevents a port from sending BPDUs.
 - Unlike BPDU Guard, it does not disable the port if it receives a BPDU.
 - Per-port: SW1(config-if)# **spanning-tree bpdupfilter enable**
 - The port will ignore any BPDUs it receives. **Use with caution!**
 - Default: SW1(config)# **spanning-tree portfast [edge] bpdupfilter default**
 - Enables BPDU Filter on all PortFast-enabled ports.
 - If the port receives a BPDU, PortFast and BPDU Filter are disabled, and it operates as a normal STP port.
 - Use **spanning-tree bpdupfilter disable** to disable it on specific ports.

